



St. Francis Institute of Technology

(Engineering College)

An Autonomous Institute, Affiliated to University of Mumbai

NAAC A+ Accredited | CMPN, EXTC, INFT NBA Accredited | ISO 9001:2015 Certified

Department of Artificial Intelligence and Machine Learning

Academic Year: 2025-2026

Class / Branch / Semester: SE - AIML IV

Term: Even (Jan. 2026 – Jun. 2026)

Course: C403-Machine Learning

ML MiniProject Ideas Based on SDGs

SDG 2 – Zero Hunger

1. **Crop Yield Prediction**
→ Rainfall, temperature → yield (Linear Regression)
2. **Soil Fertility Classification**
→ NPK values → fertile/not fertile (Decision Tree)

SDG 3 – Health and Well being

3. **Diabetes Prediction System**
→ Pima dataset (Logistic Regression/KNN)
4. **Heart Disease Risk Classifier**
→ UCI dataset (SVM/Random Forest)
5. **BMI-based Health Risk Predictor**
→ simple regression + categories

SDG 4 – Education

6. **Student Performance Prediction**
→ study hours, attendance → marks
7. **Dropout Risk Detection**
→ classify at-risk students
8. **Placement Probability Predictor**
→ CGPA + skills → placed/not placed
9. **Course Recommendation System**
→ similarity-based recommender

SDG 6 – Clean Water

10. **Water Potability Prediction**
→ safe/unsafe water classification (popular Kaggle dataset)
11. **Daily Water Usage Forecasting**
→ time series regression

SDG 7 – Clean Energy

12. **Household Energy Consumption Prediction**
13. **Solar Power Output Forecasting**
→ regression with weather data



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SDG 8 – Economic Growth

14. Loan Approval Prediction

→ classic bank dataset

15. Job Salary Prediction

→ regression on experience/skills

16. Resume–Job Matching (Basic NLP)

→ cosine similarity

SDG 11 – Sustainable Cities

17. Air Quality Index Prediction

18. Traffic Volume Prediction

19. Waste Type Classification (Dry/Wet)

→ small image dataset (simple CNN optional)

20. Driver Drowsiness Detection (camera/SVM)

eye aspect ratio, blink rate, alert system

21. Risky driving behavior detection

zig-zag steering, sudden brakes, rapid acceleration, delayed reactions, driving score

SDG 12 – Responsible Consumption

22. Food Waste Prediction in Canteen

→ forecast daily demand

23. Product Demand Forecasting

SDG 13 – Climate

24. Rainfall Prediction

25. Flood Risk Classification

SDG 15 – Life on Land

26. Plant Disease Detection (Leaf images)

→ small CNN or classical features

SDG 16 – Peace & Justice

27. Fake News Detection (Text Classification)

→ TF-IDF + Naive Bayes (very publishable)

Dr Joanne Gomes

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